

ITA14 – ETHICAL HACKING

CO1 – ASSIGNMENTS

CO1 — Foundations of Ethical Hacking

Apply the principle of the foundational concepts of ethical hacking, including types of hackers, ethical principles, legal frameworks, and the role of responsible disclosure in cybersecurity.

Introduction to Ethical Hacking, Defining Hacker, Ethics of Ethical Hacking, Ethical Hacking and the Legal System, Proper and Ethical Disclosure

S.No	Reg No	Name	Assignment Questions
1.	192421201	AKASH V	1. A company hires you to test its network but gives you only a one-line email authorization with no defined scope. Halfway through testing, you discover a critical vulnerability in a system that wasn't mentioned in the email. Do you exploit it to prove impact? Justify your answer using ethical hacking principles.
2.	192424096	AKTCHAYA A	2. You are conducting a penetration test and discover that the client's production database contains real customer financial data. Your test plan calls for exploiting a SQL injection vulnerability. How do you proceed without causing harm, and what precautions should have been taken before the engagement began?
3.	192421346	ARUN B	3. During a black-box penetration test, you accidentally trigger a denial-of-service condition on a live production server, causing an outage. Walk through what you should have done to prevent this and what your responsibility is now.
4.	192524321	Ayesha Thameema Banu M	4. A client asks you to test their systems but insists you not inform their IT department, wanting to test the team's real-world detection capability. What ethical and practical risks does this create,

			and how would you structure the engagement to protect yourself?
5.	192424312	BANDAM AKSHAY KUMAR	5. You're asked to perform a "red team" exercise versus a "vulnerability assessment." Using a real-world scenario of a bank wanting to test physical and digital security, explain how your approach, ethics, and rules of engagement would differ between the two.
6.	192521369	Dhinesh K	6. While testing a web application, you find a vulnerability that could allow you to access other clients' data hosted on the same shared server. This is outside your contracted scope. What do you do, and how does this scenario illustrate the limits of ethical hacking engagements?
7.	192524482	Eduru Rohan	7. A junior tester on your team wants to use automated exploitation tools without fully understanding what they do, "because it's faster." Using this scenario, explain the risks of tool-dependent hacking and the professional standards expected of an ethical hacker.
8.	192512277	Florace Allen Athisayaraj F	8. Your client's competitor secretly offers you money to share findings from a penetration test you conducted for the client. Analyze this scenario in terms of professional ethics, confidentiality, and potential legal consequences.
9.	192211653	GALI VENKATA MAHESWAR	9. A 16-year-old finds a flaw in their school's grading portal and quietly emails the administration instead of exploiting it. Classify this individual using standard hacker categorizations and justify your reasoning.
10	192424103	GOLAMARY NITHYA SREE	10. An employee, angry after being denied a promotion, uses their legitimate system access to leak internal documents to a journalist. Is this "hacking"? Analyze the actor's classification, motive, and the ethical/legal implications.
11	192372384	GUNDRAJULA HARSHITHA	11. A security researcher publicly releases a working exploit for a vulnerability before the vendor has issued a patch, arguing that public pressure is the only way to force a fix. Discuss whether this qualifies as

			ethical hacking, and where this person falls on the hacker spectrum.
12	192512159	Giridurgesh V	12. A nation-state-sponsored group breaches a hospital's systems to disrupt operations during a geopolitical conflict. Analyze this scenario across the categories of hacker types and explain why classification matters for response and law.
13	192524475	Hemalatha C	13. A person with no malicious intent scans random IP addresses on the internet "just to see what's out there" without permission. Argue both sides: is this person a "grey hat," and what ethical line have they crossed, if any?
14	192311057	JAYAPRAKASH V	14. Two people independently discover the same zero-day vulnerability. One reports it to the vendor; the other sells it on a dark web marketplace. Compare and contrast their classifications, motivations, and the consequences each might face.
15	192312335	KUNDETI JAYA SRI	15. A company employee is asked by management to secretly access a colleague's email account to gather evidence of misconduct, using company-owned credentials. Is this employee acting as an "ethical hacker"? Defend your position.
16	192521038	Kalpana J	16. An AI-powered bot automatically probes IoT devices for default passwords and reports vulnerable devices to a public database without owner consent. Analyze whether the creator of this bot fits any traditional hacker classification, and what new categories might be needed.
17	192525001	Kavish A	17. You're performing an authorized penetration test and discover evidence that an employee of the client company is engaged in a separate, unrelated crime (e.g., embezzlement) unrelated to your scope. What are your ethical obligations, and how do they interact with your contractual scope?
18	192521262	Kaviya K	18. A client asks you to test only certain departments, but you strongly suspect the departments excluded from scope contain the greatest security risk (possibly

			deliberately hidden by executives). Discuss the ethical tension between contractual boundaries and your professional duty to protect the organization.
19	192524393	Keshavan Kumaar S S	19. During a social engineering assessment, you successfully manipulate a receptionist into revealing sensitive information by pretending to be a grieving relative of an employee. Evaluate the ethics of this pretext, considering psychological harm versus test realism.
20	192421320	LQBAL AHMED M	20. You discover a vulnerability so severe that exploiting it, even briefly, could crash critical hospital patient-monitoring systems. Your contract requires "full exploitation to confirm impact." How do you ethically resolve this conflict between contractual obligation and patient safety?
21	192511037	Lokesh S	21. Your firm offers a discount to clients who agree to let you reuse "anonymized" data from their penetration test in a public case study or training material. Analyze the ethical issues around consent, anonymization, and confidentiality in this business model.
22	192373046	MALLURI DIVYA	22. You are asked to test the security awareness of employees by sending a phishing email that impersonates a company layoff announcement, since it's proven to get high click rates. Discuss the ethical boundaries of social engineering tests and psychological impact on staff.
23	192312472	MATTIPA BHARGAV	23. A fellow ethical hacker on your team wants to keep a "personal copy" of exploit code and access credentials discovered during a client engagement "in case it's useful later." Analyze the ethical and professional violations in this scenario.
24	192565104	Mohammed Ibrahlim M	24. You are contracted by a company to test its systems, but during recon you realize the company's core business model itself may be operating a scam that harms consumers. Do professional ethics require you to report this, and to whom, given your role is purely technical?

25	192525236	Mondepulanka Devi Srilatha	25. You are hired by a US-based company to test servers that are physically hosted in another country with different (and less protective) cybersecurity laws. Analyze the jurisdictional and legal issues you must resolve before starting the engagement.
26	192421101	NITHIYA SHREE M	26. Midway through a penetration test, the client's CEO verbally tells you to "go ahead and test that system too," but it isn't in the signed Statement of Work. Explain the legal risks of proceeding, referencing the concept of authorization and applicable computer misuse laws.
27	192524215	Nannuri Manideep	27. You discover, during an authorized test, a backdoor that appears to have been planted by a previous (unauthorized) attacker who is still actively using it. What are your legal obligations regarding evidence handling, reporting, and potential involvement of law enforcement?
28	192511028	Nilavan K	28. A client terminates your penetration testing contract mid-engagement due to a business dispute, but you still have active reverse shells and access credentials on their systems. Explain, step by step, what legal and ethical actions you must take immediately.
29	192311253	POKURU VEERASWAMI	29. An ethical hacker signs a contract with vague scope language ("test the network") and later exploits a cloud-based third-party service integrated with the client's systems. The third party threatens legal action. Analyze where liability may fall and how proper scoping and legal documentation could have prevented this.
30	192311156	PONNAPUREDDY JYOTHIKA	30. A government asks a security firm to hack into a foreign company's servers "in the national interest," without a formal legal warrant applicable across borders. Discuss the legal and ethical grey areas this creates for the hackers involved, referencing relevant legal frameworks like the Computer Fraud and Abuse Act (or equivalent).
31	192512100	Ritesh Y	31. You are asked to sign an NDA and a "get out of jail free" authorization letter

			before a physical penetration test (attempting to break into a building). During the test, local police detain you before you can produce the letter. Explain what legal protections such authorization documents provide, and their limitations in real time.
32	192524423	S Kamali	32. A company's Terms of Service prohibit "unauthorized testing" of their systems, but a security researcher argues that responsibly probing publicly accessible endpoints without causing damage should be legal under public interest research exceptions. Analyze this argument using real-world legal precedent or debate.
33	192511014	S Mohammed Sabiq	33. You discover a critical zero-day vulnerability in widely-used medical device software during an authorized test. The vendor is unresponsive for 90 days despite multiple attempts to contact them. Using a responsible disclosure framework, explain your next steps and justify the timeline.
34	192324303	SANTHOSH N R	34. A security researcher publicly tweets details of a vulnerability in a popular banking app within hours of discovering it, without contacting the bank first, citing "the public has a right to know." Analyze this against standard disclosure practices and the potential real-world consequences.
35	192324139	SIRIYALA MUBINA	35. You report a vulnerability to a company through their official bug bounty program, but the company threatens legal action against you instead of paying the promised reward, claiming you "exceeded authorized scope." Discuss the disclosure and legal risks researchers face, and how to mitigate them beforehand.
36	192572139	Sharmila M	36. During a coordinated disclosure process, the vendor asks you to delay public disclosure indefinitely with no fixed patch timeline, citing "business reasons" unrelated to the technical fix. As the researcher, how do you balance ethical

			disclosure obligations against vendor pressure?
37	192521067	T S Harini	37. You find a vulnerability affecting an open-source library used by thousands of downstream applications. Disclosing to the maintainer alone won't reach all affected parties quickly. Design a disclosure strategy that balances speed, safety, and coordination across the ecosystem.
38	192424316	VANGIMALLA BALA SIVA PRASAD REDDY	38. A company offers you a "hush payment" larger than their bug bounty reward, on the condition you never disclose the vulnerability, even after it's patched. Analyze the ethical implications of accepting this offer versus standard responsible disclosure.
39	192372113	VUMMADI PALLAVI	39. You discover a vulnerability while testing one client's system, but the flaw actually exists in a third-party software product used by thousands of other organizations who are not your clients. Explain your ethical and practical disclosure obligations in this multi-party scenario.
40	192511030	Vijay A	40. A rival security researcher discloses a vulnerability you found first, taking public credit, after you had already privately reported it to the vendor under an embargo. Discuss the ethical norms around disclosure credit, embargoes, and how researchers should handle disputes like this professionally.

26) Midway through a penetration test, the client's CEO verbally tells you to "go ahead and test the system too", but it isn't in the signed statement of work. Explain the legal risks of proceeding, referring the legal risks concept of authorization and applicable computer misuse laws.

Proceeding with the test could be legally risky, even if the CEO gives verbal permission.

The statement of work (SOW) is the formal document that defines the scope of the penetration test, including which systems are authorized for testing. If the additional system is not listed in the signed SOW, there is no documented authorization to test it.

Legal risks include:

Lack of documented authorization: Verbal approval is difficult to prove if a dispute arises. Written authorization is the standard evidence that

the testing was permitted.

Violation of Computer misuse law: In many countries, accessing or testing a computer system without proper authorization may violate laws such as the Computer Fraud and Abuse Act (CFAA) in the United States or the Computer Misuse Act 1990 in the United Kingdom.

Civil Liability: If the testing causes downtime, database, or service disruption, the client could claim that the tester acted outside the agreed scope and seek damages.

Professional and contractual consequences: Operating beyond the signed agreement may breach the contract, damage the tester's reputation and create legal or insurance issues.

Best practice:

Before testing the additional system, stop and request a written amendment or

change order to the statement of work that clearly adds the new system to the authorized scope. Once the updated document is approved and signed by the appropriate authority, the penetration test can proceed legally and professionally.

Conclusion:

Even if the CEO verbally approves the test, you should not proceed until the authorization is documented in writing. Written authorization protects both the client and the penetration tester and helps ensure compliance with applicable computer misuse laws.